

Gustavo Alvarado

R&D Engineer — Unmanned Aerial Systems, Embedded Systems & Computer Vision
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SUMMARY

Mechatronics engineer (ESPOL) with 4 years of experience building production systems for unmanned aerial vehicles: real-time computer-vision pipelines, low-latency Qt desktop applications, and embedded Linux telemetry services. Comfortable owning a system end to end — architecture, implementation, and deployment onto constrained hardware. Hands-on UAS field experience and a remote-pilot certificate endorsed by DGAC.

EXPERIENCE

Robiotec Ecuador S.A.

Oct 2022 — Present

Guayaquil, Ecuador

R&D Engineer — Drone Specialist & Software Developer

Apr 2024 — Present

- Sole architect and developer of the operator application for a high-end, military-use surveillance camera (**Python**, **PyQt6**): full camera control through the vendor SDK driven by professional joysticks, live video client, configuration and deployment. Rebuilt the video front end on FFmpeg (**PyAV**), cutting client-side buffering on a **1080p** feed served remotely over a **satellite link**.
- Took a **YOLOv8** mineral-detection system from prototype to production on **NVIDIA Jetson**: reworked real-time scheduling and actuator triggering and converted inference to **TensorRT** so detection lands inside the actuators' timing window. Also configured a real-time kernel on the Jetson.
- Redesigned and rewrote the onboard telemetry service running on **Raspberry Pi** under embedded GNU/Linux (**Python**, **SurrealDB**), prioritising critical parameters over a direct link to the flight controller, with a scalable local store.
- Designed the end-to-end architecture of a satellite-linked UAS control system integrating a high-end surveillance payload.
- Built a flight-log analytics pipeline over **500k+** telemetry records (**Polars**, **Plotly**, **marimo**) quantifying energy consumption per flight phase, with sub-second queries replacing a pandas/Matplotlib/Jupyter workflow.
- Adapted **ArduPilot** control loops to a new 3D-printed flight controller and a new sensor set for a fixed-wing glider, and scripted hexacopter flight logic in **Lua**.
- Rebranded an open-source UAS control suite for corporate deployment (**C++/Qt**, **Docker**, **Bash**) under full licence compliance.
- Pilot in command for VTOL fixed-wing operations: route planning, calibration and mission configuration.
- Maintained the 3D-printing equipment and designed mounts and work accessories in Autodesk Fusion.

R&D Lab Assistant

Apr 2023 — Mar 2024

- Wrote the **Python** integration layer between the **MAVLink** protocol and a partner company's power-supply system.
- Built the backend of a power-monitoring system for industrial mills over **Modbus TCP (Node-RED/JavaScript on Linux)**.
- Enabled a **93 km** unsupervised point-to-point fixed-wing VTOL mission — GNSS/GPS integration, radio-link revision and geolocation-software updates, validated in flight, payload and comms testing.
- Ran CFD simulations (Autodesk Inventor) on a tandem-wing drone airframe.
- Installed and supported Linux environments for computer-vision projects.
- Researched and developed the communication system between radio control and propulsion for an aquatic drone (Arduino C).

R&D Lab Intern

Oct 2022 — Mar 2023

- Built the backend of a satellite geolocation service consuming **Iridium SBD** messages through a webhook, persisted in **SQLite (Node-RED/JavaScript)**.
- Shipped a **Qt5/Python** colour-picker front end feeding image-region selections into a computer-vision colour-recognition service.
- Configured ArduPilot flight control for electric VTOL fixed-wing and high-payload aquatic platforms.

SELECTED PROJECTS

Low-latency AI inference pipeline

Python · YOLOv8 · TensorRT · CUDA · NVIDIA Jetson · Linux RT kernel · uv

Production inference path for real-time mineral detection on NVIDIA Jetson: TensorRT conversion, real-time kernel configuration and a redesigned real-time scheduling model that drives physical actuators within their timing window.

Camera control & video client for a military-use surveillance camera

Python · PyQt6 · PyAV (FFmpeg) · vendor SDK · joystick HID · satellite video link

Complete operator application for a high-end surveillance camera, built solo: camera control via the vendor SDK over professional joysticks, live video client, configuration and deployment. Video front end rebuilt on FFmpeg (PyAV) to minimise client-side buffering on a 1080p feed delivered over a satellite link.

Flight-log analytics dashboard

Python · Polars · Plotly · marimo · uv

Interactive dashboard over **500k+** telemetry records visualising voltage, current and energy consumption across drone flight phases, with sub-second query performance.

Satellite geolocation backend

Node-RED · JavaScript · SQLite · Iridium SBD · Git

Position-tracking service for long-range drones built on satellite Short Burst Data with spatial querying and local persistence.

TECHNICAL SKILLS

Languages: Python, C/C++ (embedded & Qt), JavaScript, Lua, SQL, Bash
ML & Computer Vision: PyTorch, Ultralytics YOLOv8, TensorRT, CUDA, OpenCV, Pylon (Basler)
Robotics: ROS / ROS 2, NVIDIA Jetson AGX Orin, MAVLink
Data: Polars, pandas, NumPy, Plotly, Matplotlib/Seaborn, marimo, Jupyter
Applications & UI: PyQt6 / PyQt5, Qt (C++)
Embedded & Systems: GNU/Linux (Ubuntu, Arch, NixOS), low-latency kernels, Raspberry Pi, Orange Pi, ESP32 / ESP-IDF, Teensy, Arduino, MAVLink, Modbus TCP, Node-RED
Data stores: SQLite, SurrealDB
Tooling & Infrastructure: Git, Docker, Podman, Proxmox, uv
UAS: ArduPilot, Betaflight, mission planning, VTOL fixed-wing & multirotor operations
CAD & Simulation: Autodesk Fusion, Inventor, Inventor CFD, AutoCAD, MATLAB, Proteus, MPLAB X

EDUCATION & CERTIFICATIONS

B.Eng. in Mechatronics Engineering	2023
<i>ESPOL — Escuela Superior Politécnica del Litoral — Guayaquil, Ecuador</i>	
• Teaching assistant — Introduction to Data Science; Programming Fundamentals. • Participant, Pyweeknd hackathon.	
Remote Pilot Certificate — Multirotor, VLOS	2025
<i>Aeronaut Systems, endorsed by DGAC Ecuador — Guayaquil, Ecuador</i>	

LANGUAGES

Spanish: Native
English: B2 — professional working proficiency

REFERENCES

Available on request.